Keerti Vijay Ananth

+91 91594 44964 | keertivijayananthprof@gmail.com | linkedin.com/in/keerti-vijay-ananth | github.com/kvijay0611

Experience

Motorola Solutions *Software Engineer Intern* Feb 2026 – Oct 2026

Project 1: POC on Fault-Tolerant Messaging for Mission-Critical Communications

- Contributed to the system design and testing of a **fault-tolerant, multi-data-center messaging architecture** for critical communications, using an **N+1 redundancy model** so the remaining data centers absorb the full load if one site fails, with no dropped signals or service interruption.
- Evaluated **Temporal and NATS** for reliable message delivery, workflow orchestration and horizontal scaling, to identify the best fit for the next-generation production system.
- Designed workflows that simulated data-center failures, validating failover behavior and enabling **zero-downtime deployments** for critical communications, to guide architectural decisions.

Project 2: RabbitMQ High Availability

- Investigated how **RabbitMQ Shovel** could be integrated into an existing production architecture to improve availability and prevent message loss across sites, and documented the design trade-offs.
- Tested next-generation availability requirements across **10+ failure scenarios**, load-testing **50K+ messages per node** on a local setup to confirm critical connections stayed up and in-flight messages were preserved during failover.

BVK Exports - Textile Company *Machine Learning Intern* May 2025 – July 2025

- Developed and trained a **YOLOv8n-based fabric defect detection model** on **10,000+ labeled fabric images**, targeting automated localization of weaving defects, holes, and stains for textile quality inspection.
- Evaluated model performance on a held-out test set, achieving **85% precision, 80% recall, and 84% mAP@50** in the illustrative scenario; analyzed class-wise errors to identify missed defects and reduce false-positive predictions.
- Built a **defect analytics workflow** to quantify defect patterns and model predictions, with a hypothetical pilot evaluation indicating **20% lower manual inspection time** for the tested workflow.

Projects

TrialLens – Clinical Trial Intelligence Tool | Live Demo

- Engineered a **5-step LangGraph workflow** routing natural-language queries across SQL lookups, keyword-based document retrieval, and FDA adverse-event APIs, using **Llama 3.3 70B (Groq)** for intent parsing and cited response synthesis.
- Integrated **ClinicalTrials.gov and openFDA FAERS APIs**, implementing full-text search fallbacks for condition matching and resolving production API access issues caused by TLS fingerprinting and WAF restrictions.
- Optimized production deployment by replacing a **Chroma + sentence-transformers** retriever with a lightweight TF-IDF-style scorer, eliminating the ML runtime and resolving **512MB OOM failures**, while deploying the Dockerized service on Render with provider-agnostic LLM support.

Advanced Diabetes Prediction System

- Engineered a supervised ML pipeline using Scikit-Learn to classify diabetes risk levels from clinical and lifestyle attributes with an accuracy of **94.5%**.
- Developed modular Python scripts for feature engineering, model training, hyperparameter tuning, and performance validation.
- Integrated real-world inference through a React + Node.js interface backed by AWS SageMaker, enabling instant risk-level predictions and actionable health insights.

Education

Vellore Institute of Technology Bhopal, India

B.Tech (CSE) Health Informatics - 8.65 CGPA *Sep 2022 – Oct 2026*

Technical Skills

Languages: Python, Java, Go, SQL
AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision (YOLO), NLP, LLMs, RAG, Scikit-Learn
Data & Analytics: SQL, MySQL, Pandas, NumPy, Matplotlib, Feature Engineering, Statistical Analysis, Power BI, Tableau
Backend & Distributed Systems: Message Queues, REST APIs, RabbitMQ, NATS, Temporal, CI/CD
Cloud & Tools: AWS (SageMaker), Docker, Linux, Git, GitHub, Node.js

Co-curricular & Achievements

- Presented the paper "**Analysis of Nanomaterial-Based Biosensor Health Monitoring Systems with Artificial Intelligence** for Smart Healthcare" at **ICINVENTS-25, SKCET**.
- Assisted in making of **Python Tamil E-Content Course** at VIT Bhopal Nov 2023